Erfan Saydanzad

Assistant Professor of Physics, Department of Physics, Kennesaw State University

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RESEARCH

Current

Ultrafast and strong-field physics
Theoretical and computational
Multi-channel strong-field ionization and photoelectron emission from metallic, dielectric and semiconducting nanoparticles; attosecond plasmonic near-field imaging; time-domain Fano theory of autoionizing molecular Rydberg states and nonadiabatic dynamics.

APPOINTMENTS

Aug 2025 –

Assistant Professor of Physics
Kennesaw State University, Marietta, GA
Department of Physics

Dec 2021 – Jul 2025

Postdoctoral Research Associate
J. R. Macdonald Laboratory, Kansas State University, KS

2013–2015, 2021–2022

Lecturer and Laboratory Instructor
Kansas State University, KS

Aug 2009 – Aug 2013

Lecturer and Laboratory Instructor
University of Kurdistan, Iran

2011–2012

Researcher, Organic LED Laboratory
K. N. Toosi University of Technology, Iran

EDUCATION

2013–2021

Ph.D., Atomic, Molecular and Optical Physics
Kansas State University, Manhattan, KS, USA

2006–2009

M.S., Photonics
Graduate University of Advanced Technology, Kerman, Iran

2002–2006

B.S., Solid-State Physics
University of Kurdistan, Iran

TEACHING AT KENNESAW STATE

PHYS 3260

Mathematical Physics
Kennesaw State University
Lecture notes written and published openly.

PHYS 2211

Principles of Physics I
Kennesaw State University

PHYS 2211L

Introductory Physics Laboratory I
Kennesaw State University

Quantum Mechanics and Spectroscopy

Part of Direct Method, guided undergraduate research
Kennesaw State University
One-to-one research mentoring for undergraduates.

PUBLICATIONS

1. E. **Saydanzad**, J. Powell, T. Renner, A. Summers, D. Rolles, C. Trallero-Herrero, M. F. Kling, A. Rudenko, and U. Thumm, "Generation of fast photoelectrons in strong-field emission from metal nanoparticles," *Nanophotonics* **14**, 1355–1364 (2025).
2. E. **Saydanzad**, J. Powell, A. Summers, S. J. Robatjazi, C. Trallero-Herrero, M. F. Kling, A. Rudenko, and U. Thumm, "Enhanced cutoff energies for direct and rescattered strong-field photoelectron emission of plasmonic nanoparticles," *Nanophotonics* **12**, 1931–1942 (2023).
3. J. A. Powell, J. Li, A. Summers, S. J. Robatjazi, M. Davino, P. Rupp, E. **Saydanzad**, C. M. Sorensen, D. Rolles, M. F. Kling, and others, "Strong-field control of plasmonic properties in core–shell nanoparticles," *ACS Photonics* **9**, 3515–3521 (2022).
4. E. **Saydanzad**, J. Li, and U. Thumm, "Strong-field ionization of plasmonic nanoparticles," *Physical Review A* **106**, 033103 (2022).
5. J. Li, E. **Saydanzad**, and U. Thumm, "Imaging plasmonic fields with atomic spatiotemporal resolution," *Physical Review Letters* **120**, 223903 (2018).
6. E. **Saydanzad**, J. Li, and U. Thumm, "Spatiotemporal imaging of plasmonic fields near nanoparticles below the diffraction limit," *Physical Review A* **98**, 063422 (2018).
7. J. Li, E. **Saydanzad**, and U. Thumm, "Attosecond time-resolved streaked photoelectron spectroscopy of transition-metal nanospheres," *Physical Review A* **95**, 043423 (2017).
8. E. **Saydanzad**, J. Li, and U. Thumm, "Characterization of induced nanoplasmonic fields in time-resolved photoemission: a classical trajectory approach applied to gold nanospheres," *Physical Review A* **95**, 053406 (2017).
9. J. Li, E. **Saydanzad**, and U. Thumm, "Retrieving plasmonic near-field information: a quantum-mechanical model for streaking photoelectron spectroscopy of gold nanospheres," *Physical Review A* **94**, 051401(R) (2016).
10. M. H. Rezvani, F. Farajollahi, A. Nikfarjam, P. Bakhtiarpour, and E. **Saydanzad**, "Effect of solvents, their mixture and thermal annealing on the performance of solution processed polymer light-emitting diodes," *Materials* **6**, 1994–2006 (2013).

PREPRINTS

1. E. **Saydanzad**, V.-H. Hoang, and U. Thumm, "Time-resolved Rydberg dynamics in molecules," *arXiv:2610.27286 [physics.atom-ph]* (2026) [Submitted to Physical Review A].

MANUSCRIPTS IN PREPARATION

1. T. Renner, J. Powell, et al., A. Rudenko, U. Thumm, and E. **Saydanzad**, "Breakdown of ponderomotive cutoff energy scaling in strong-field emission from dielectric nanoparticles."